

The Continuous-Time Fourier Transform (CTFT)

CTFT

Continuous-Time Fourier Transform (CTFT)

For continuous-time aperiodic signals

- Synthesis equation:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

- Analysis equation:

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

CTFT

Cyclic frequency can also be used.

- ▶ Synthesis equation:

$$x(t) = \int_{-\infty}^{\infty} X(F) e^{j2\pi Ft} dF$$

- ▶ Analysis equation:

$$X(F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt$$

CTFT: Dirichlet Conditions

CTFT: Dirichlet Conditions

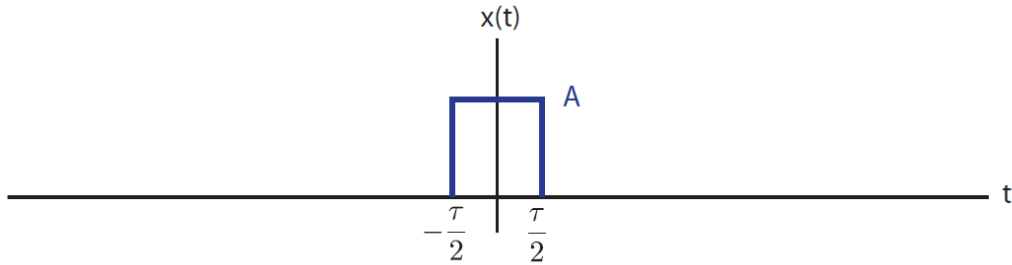
- ▶ Allowing $T_p \rightarrow \infty$ in CTFS Dirichlet conditions:
 1. $x(t)$ has a finite number of finite discontinuities.
 2. $x(t)$ has a finite number of maxima and minima.
 3. $x(t)$ is absolutely integrable:

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

CTFT Example

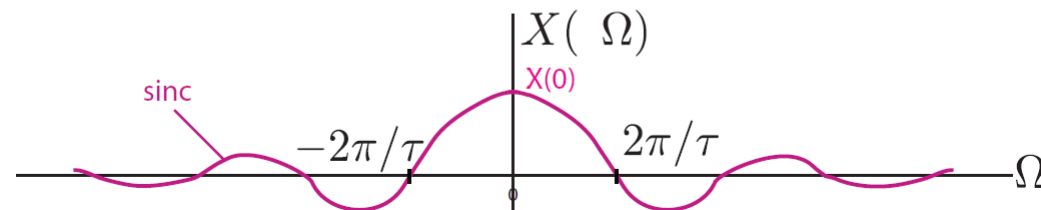
CTFT: Example

Find the CTFS of the following periodic square wave:



$$\begin{aligned} X(\Omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt = \int_{-\tau/2}^{\tau/2} A e^{-j\Omega t} dt \\ &= A \left. \frac{e^{-j\Omega t}}{-j\Omega} \right|_{-\tau/2}^{\tau/2} = 2A \frac{\sin(\Omega\tau/2)}{\Omega} \end{aligned}$$

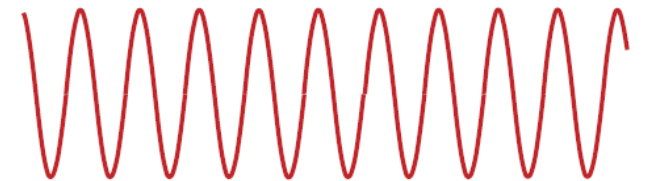
$$X(\Omega) = 2A \frac{\sin(\Omega\tau/2)}{\Omega}$$



CTFT Intuition

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

- ▶ We may consider $x(t)$ as a linear combination of $e^{j\Omega t}$ for $\Omega \in \mathbb{R}$.
- ▶ The larger $|X(\Omega)|$, the more $x(t)$ will look like a sinusoid with Ω .



CTFT Duality

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$
$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$\begin{array}{ccc} x(t) & \xleftrightarrow{\mathcal{F}} & X(\Omega) \\ \text{rectangle} & \xleftrightarrow{\mathcal{F}} & \text{sinc} \\ \text{sinc} & \xleftrightarrow{\mathcal{F}} & \text{rectangle} \\ \text{convolution} & \xleftrightarrow{\mathcal{F}} & \text{multiplication} \\ \text{multiplication} & \xleftrightarrow{\mathcal{F}} & \text{convolution} \end{array}$$

CTFT: Magnitude and Phase

Q: Which is more **important** for a given signal?

- ▶ Does one component (magnitude or phase) contain more **information** than another?
- ▶ When filtering, if we had to **preserve** one component (magnitude or phase) more, which one would it be?

CTFT: Audio Signal

- ▶ An audio signal is represented by a real function $x(t)$.
- ▶ The function $x(-t)$ represents playing the audio signal backwards.
- ▶ Since $x(t)$ is real:

$$\begin{aligned} X(\Omega) &= X^*(-\Omega) \\ |X(\Omega)| &= |X^*(-\Omega)| = |X(-\Omega)| \quad \text{since } |c| = |c^*| \text{ for } c \in \mathbb{C} \end{aligned}$$

- ▶ Therefore,

$$|X(\Omega)| = |X(-\Omega)|$$

That is, the CTFT magnitude is **even** for $x(t)$ **real**.

CTFT: Audio Signal

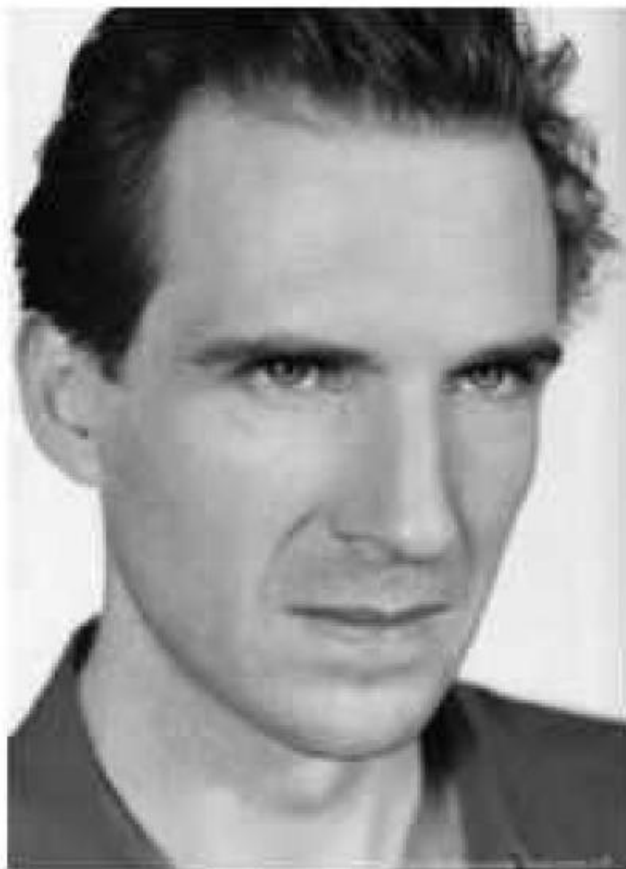
► Recall, $x(t) \xleftrightarrow{\mathcal{F}} X(\Omega)$ $x(-t) \xleftrightarrow{\mathcal{F}} X(-\Omega)$

► Therefore,

$$\underbrace{|X(\Omega)|}_{\text{spectrum magnitude of } x(t)} = \overbrace{|X(-\Omega)|}^{\text{spectrum magnitude of } x(-t)}$$

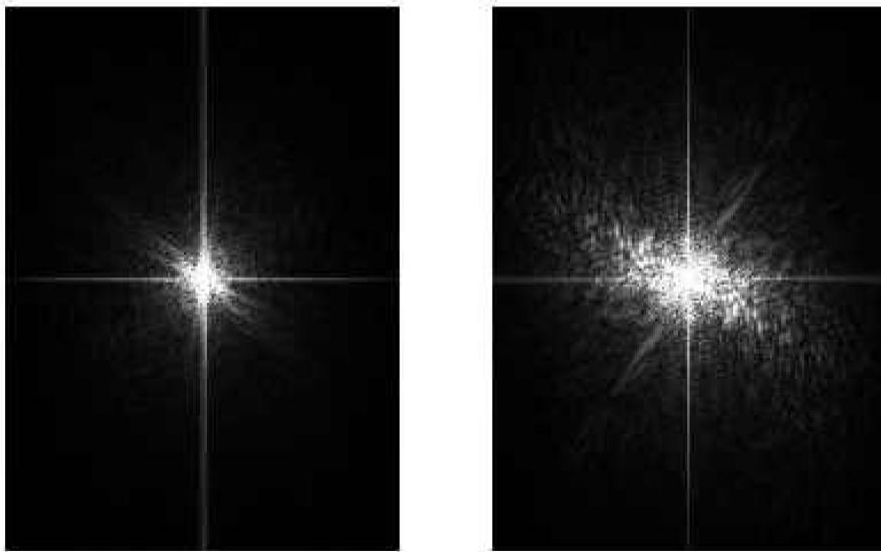
Therefore, the magnitude of the FT of an audio signal played forward and backward is the same!

Example:
Still Image,
 $x(t_1, t_2)$

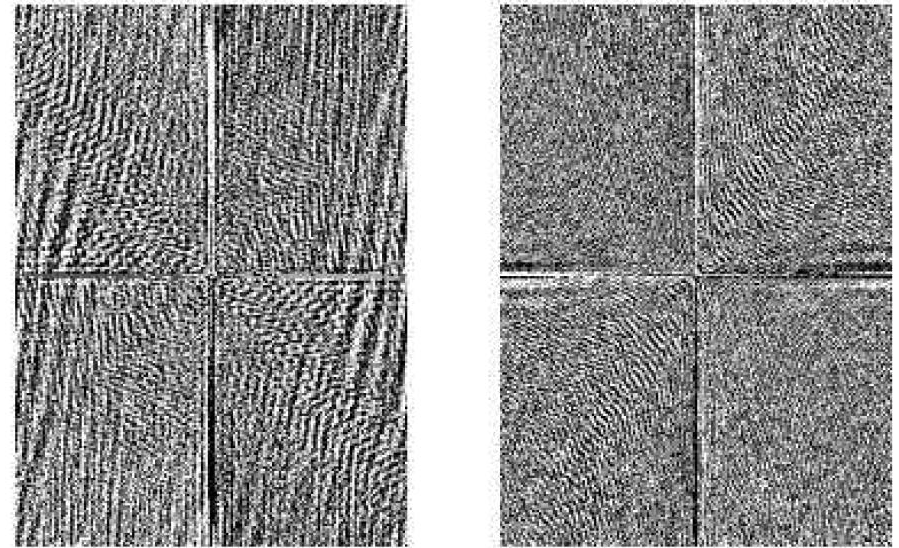


Magnitude and Phase

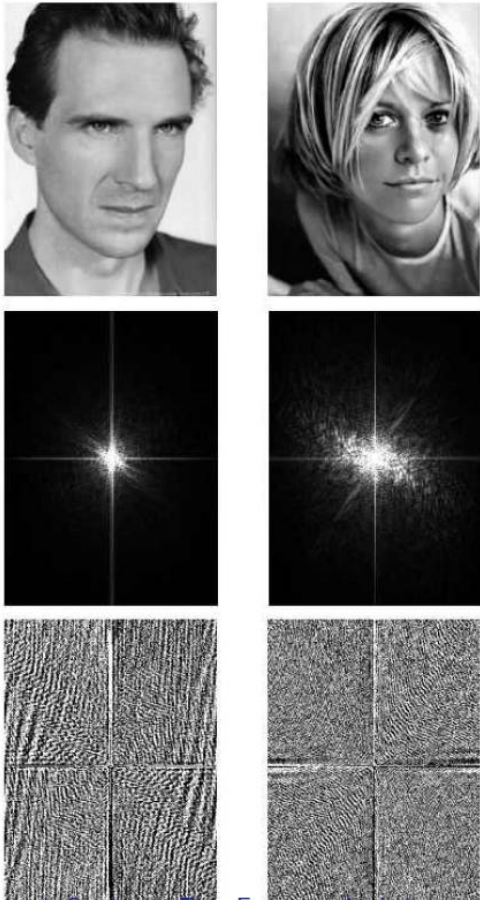
Example: $|X(\Omega_1, \Omega_2)|$



Example: $\angle X(\Omega_1, \Omega_2)$



Reconstruction

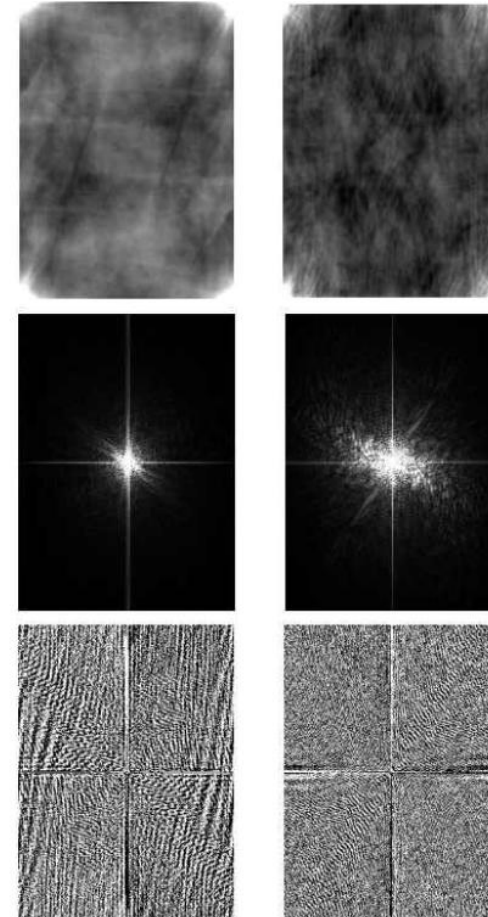


Reconstruction using

magnitude only

Top Left Photo: Ralph's
magnitude is the same,
Phase = 0

Top Right Photo: Meg's
magnitude is the same,
Phase = 0

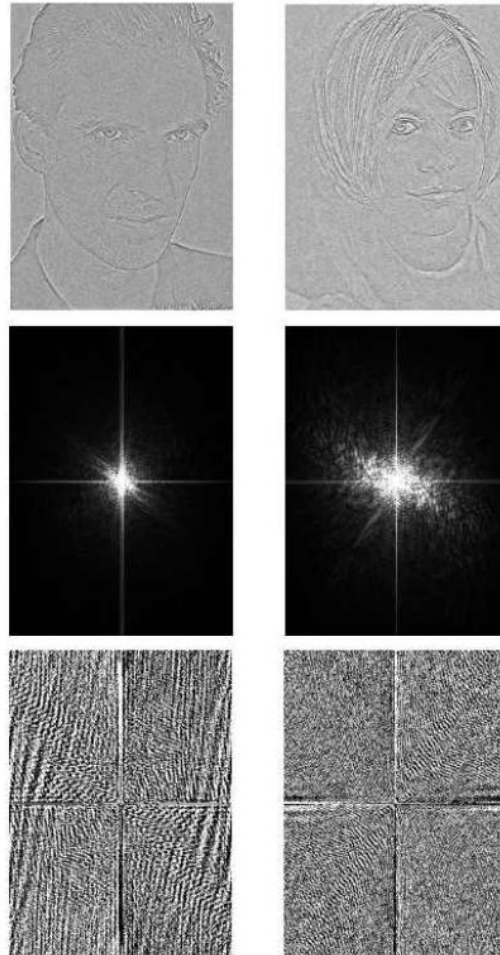


Reconstruction

Reconstruction using
phase only

Top Left Photo: Ralph's
magnitude normalized to
one, Phase is the same

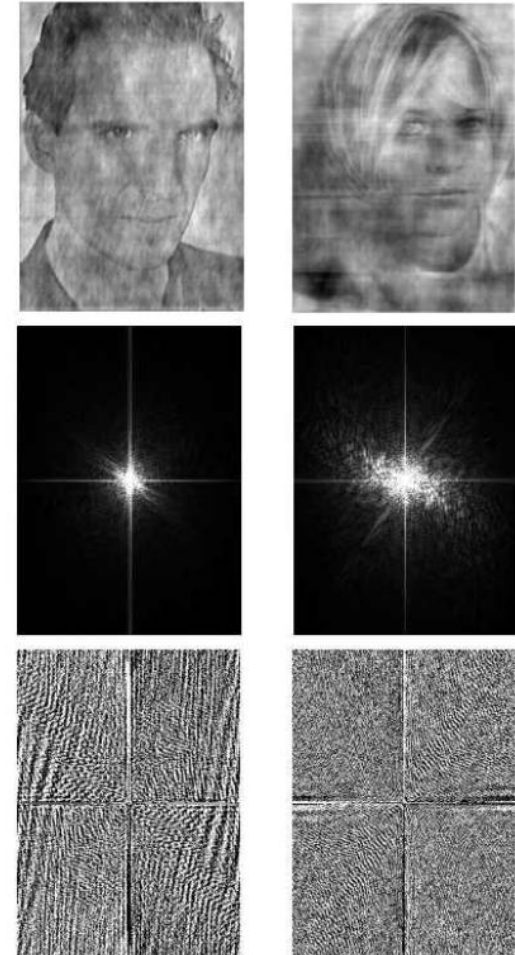
Top Right Photo: Meg's
magnitude normalized to
one, Phase is the same



Reconstruction swapping
magnitude and **phase**
of the images.

Top Left Photo: Ralph's
phase + Meg's magnitude

Top Right Photo: Meg's
phase + Ralph's magni-
tude



CTFT: Magnitude and Phase

Q: Which is more **important** for a given signal? **A: Phase**

- ▶ Does one component (magnitude or phase) contain more **information** than another? **A: Yes, phase**
- ▶ When filtering, if we had to **preserve** one component (magnitude or phase) more, which one would it be?
A: Phase

Frequency Analysis of Discrete-Time Signals

The Fourier Series for Discrete-Time Periodic Signals

Synthesis:

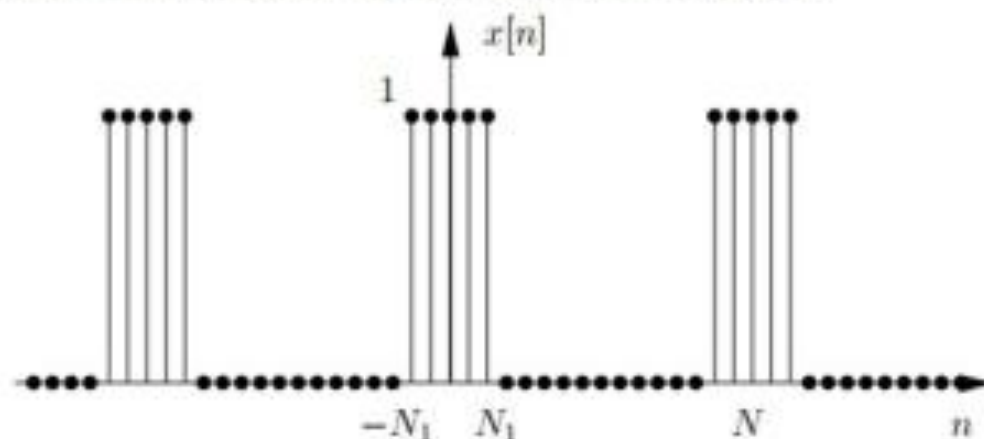
$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N}$$

Analysis:

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

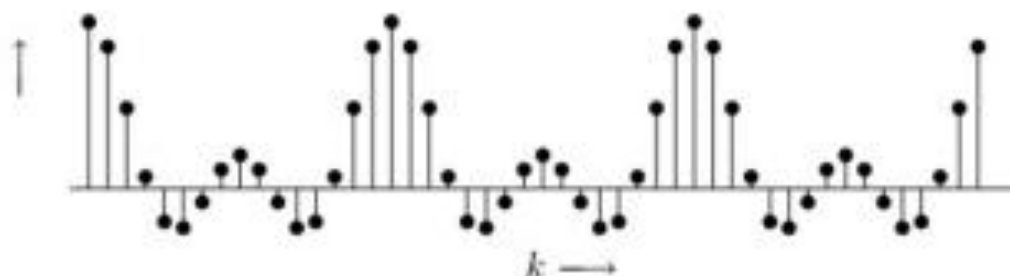
- ▶ Sequence $x(n)$ with period N , $x(n) = x(n + N)$
- ▶ $c_k = c_{k+N}$, c_k is a periodic sequence with fundamental period N
- ▶ For a sampling frequency F_s ; range $0 \leq k \leq N - 1$ corresponds to $0 \leq F \leq F_s$

- Discrete-time periodic square wave with period N



- Fourier series coefficients

$$\begin{aligned}
 &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk(2\pi/N)n} \\
 &= \begin{cases} \frac{1}{N} \frac{\sin(2\pi k(N_1 + 1/2)/N)}{\sin(\pi k/N)}, & k \neq 0, \pm N, \pm 2N, \dots \\ \frac{2N_1 + 1}{N}, & k = 0, \pm N, \pm 2N, \dots \end{cases}
 \end{aligned}$$



The Fourier Series for Discrete-Time Periodic Signals

Power Density Spectrum of Periodic Signals

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

in terms of Fourier coefficients $\{c_k\}$

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x(n)x^*(n) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) \left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N} \right)$$

$$P_x = \sum_{k=0}^{N-1} c_k^* \left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right]$$

$$P_x = \sum_{k=0}^{N-1} |c_k|^2$$

The Fourier Series for Discrete-Time Periodic Signals

Real signals

If $x(n)$ is a real signal, $x^*(n) = x(n)$, is shown that $c_k^* = c_{-k}$ i.e. spectrum $c_k, k = 1, 2, \dots, N/2$ completely describes the signal in the frequency domain.

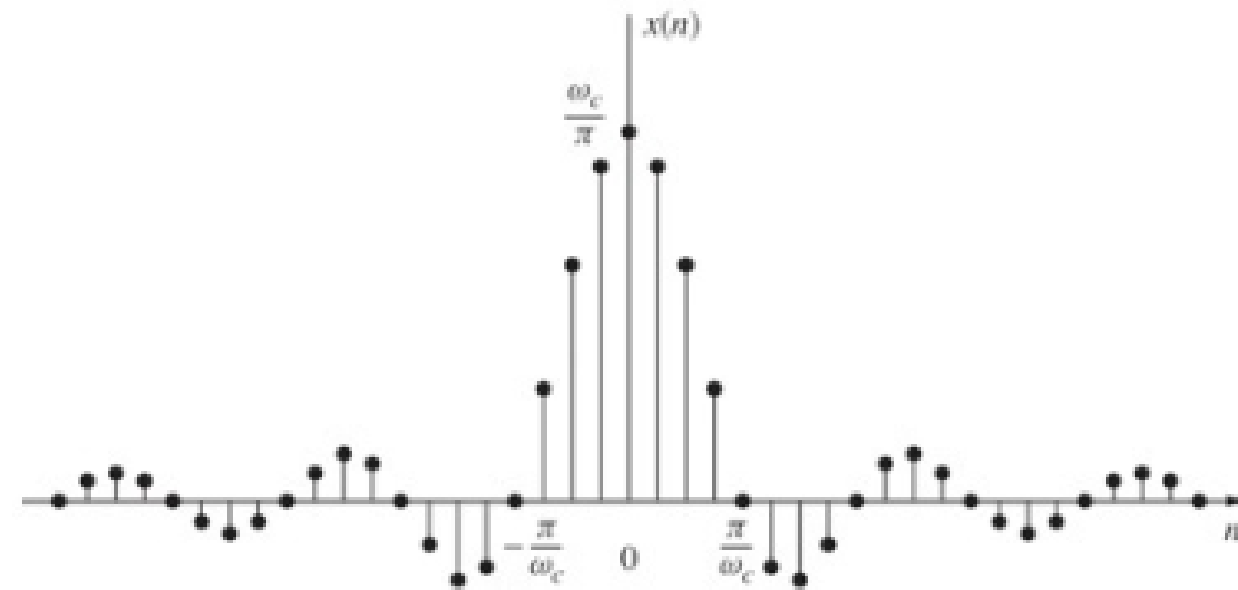
The Fourier Series for Discrete-Time Aperiodic Signals

Synthesis Equation:

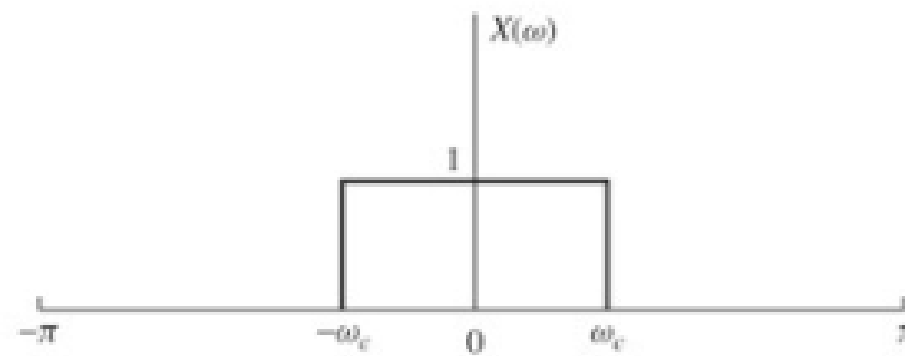
$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$$

Analysis Equation:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$



(a)



(b)

Figure 4.2.4 Fourier transform pair in (4.2.35) and (4.2.36).

The Fourier Series for Discrete-Time Aperiodic Signals

Energy Density Spectrum of Aperiodic Signals

Recall energy of a discrete-time signal $x(n)$

$$\begin{aligned} E_x &= \sum_{n=-\infty}^{\infty} |x(n)|^2 \\ E_x &= \sum_{n=-\infty}^{\infty} x^*(n)x(n) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) e^{-j\omega n} d\omega \right] \\ E_x &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right] d\omega \\ E_x &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \end{aligned}$$

- Energy Density Spectrum of $x(n)$: $S_{xx}(\omega) = |X(\omega)|^2$

The Fourier Series for Discrete-Time Aperiodic Signals

Real signals

Let $x(n)$ be a real signal:

- ▶ $|X(-\omega)| = |X(\omega)|$, even symmetry
- ▶ $S_{xx}(-\omega) = S_{xx}(\omega)$, even symmetry

⇒ Frequency range of real discrete-time signals can be limited to
 $0 \leq \omega \leq \pi$ or $0 \leq F \leq F_s/2$

Relationship of the Fourier Transform to the Z-Transform

Recall

z-Transform of sequence $x(n)$:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}, \text{ ROC: } r_2 \leq |z| \leq r_1$$

Express complex z in polar format:

$$z = re^{j\omega} \Rightarrow X(z)|_{z=re^{j\omega}} = \sum [x(n)r^{-n}] e^{-j\omega n}$$

If $X(z)$ converges for $|z| = 1$:

$$X(z)|_{z=e^{j\omega}} = X(\omega) = \sum [x(n)r^{-n}] e^{-j\omega n}$$

- Fourier Transform is viewed as the z-Transform of the sequence evaluated on the unit circle.

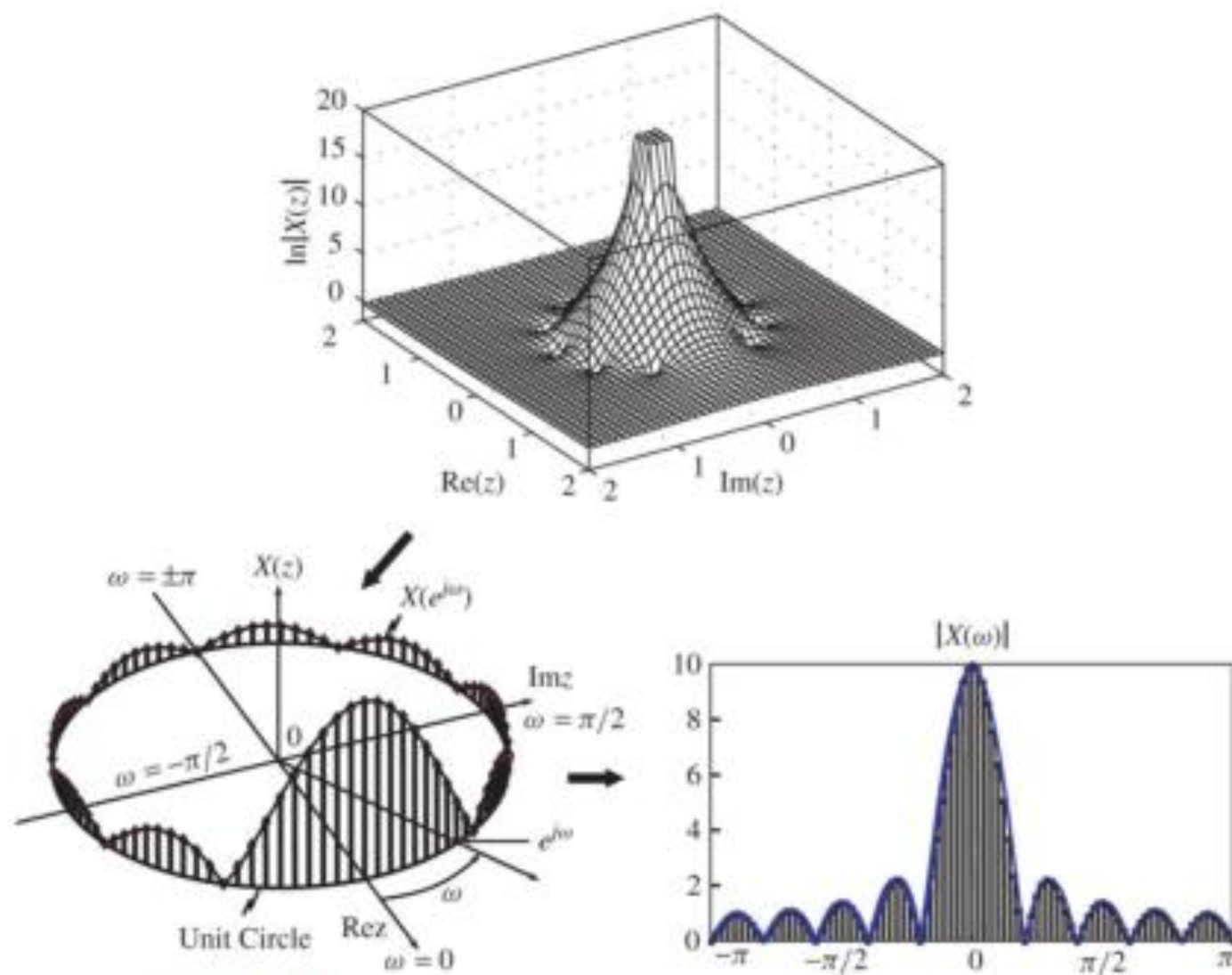


Figure 4.2.9 relationship between $X(z)$ and $X(\omega)$ for the sequence in Example 4.2.4, with $A = 1$ and $L = 10$

Figure courtesy of Proakis and Manolakis, Digital Signal Processing.

Frequency-Domain classification of Signals: Bandwidth

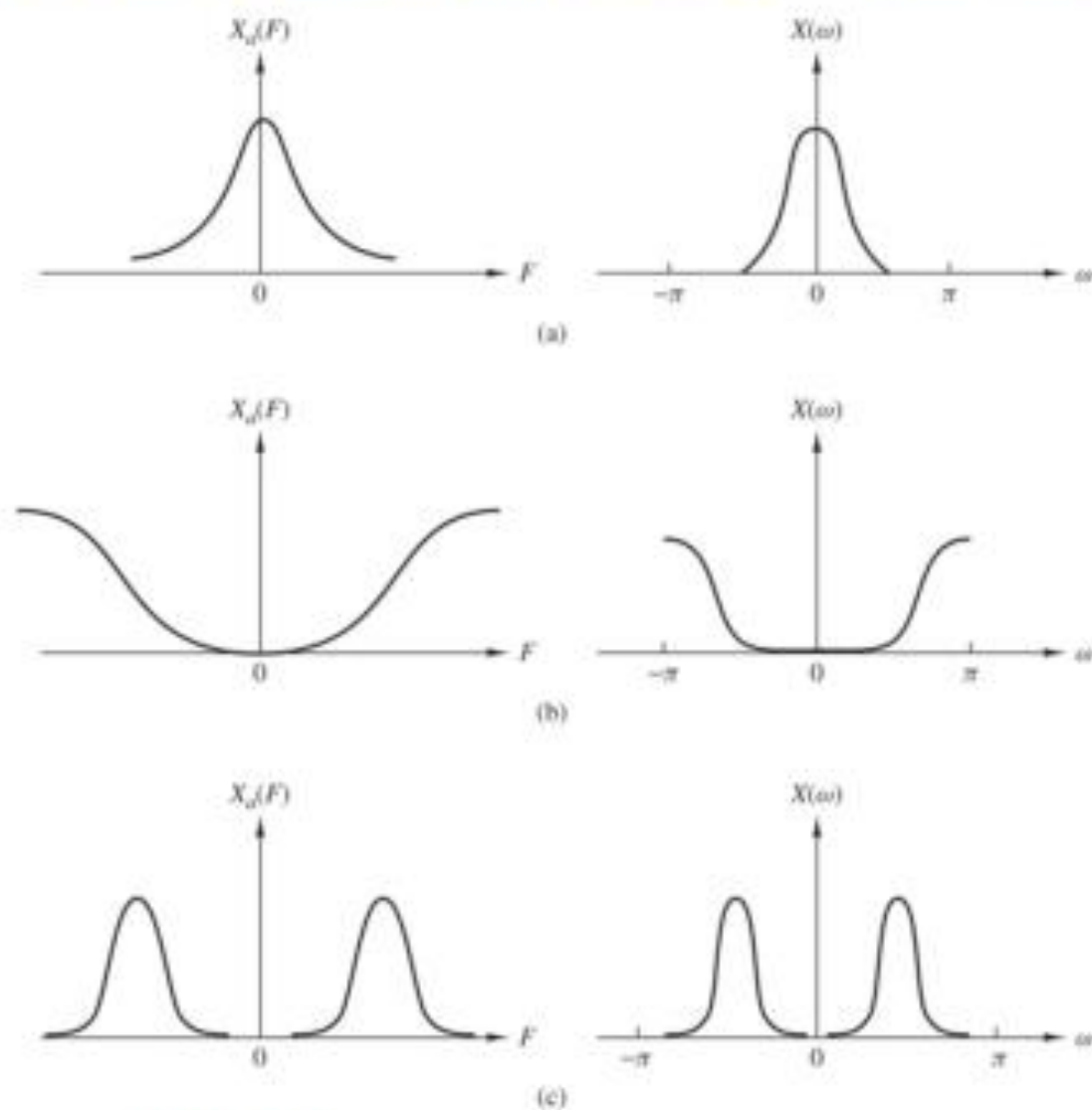


Figure 4.2.10 (a) Low-frequency, (b) high-frequency, and (c) medium-frequency signals.

Frequency-Domain classification of Signals: Bandwidth

Broad frequency domain classification

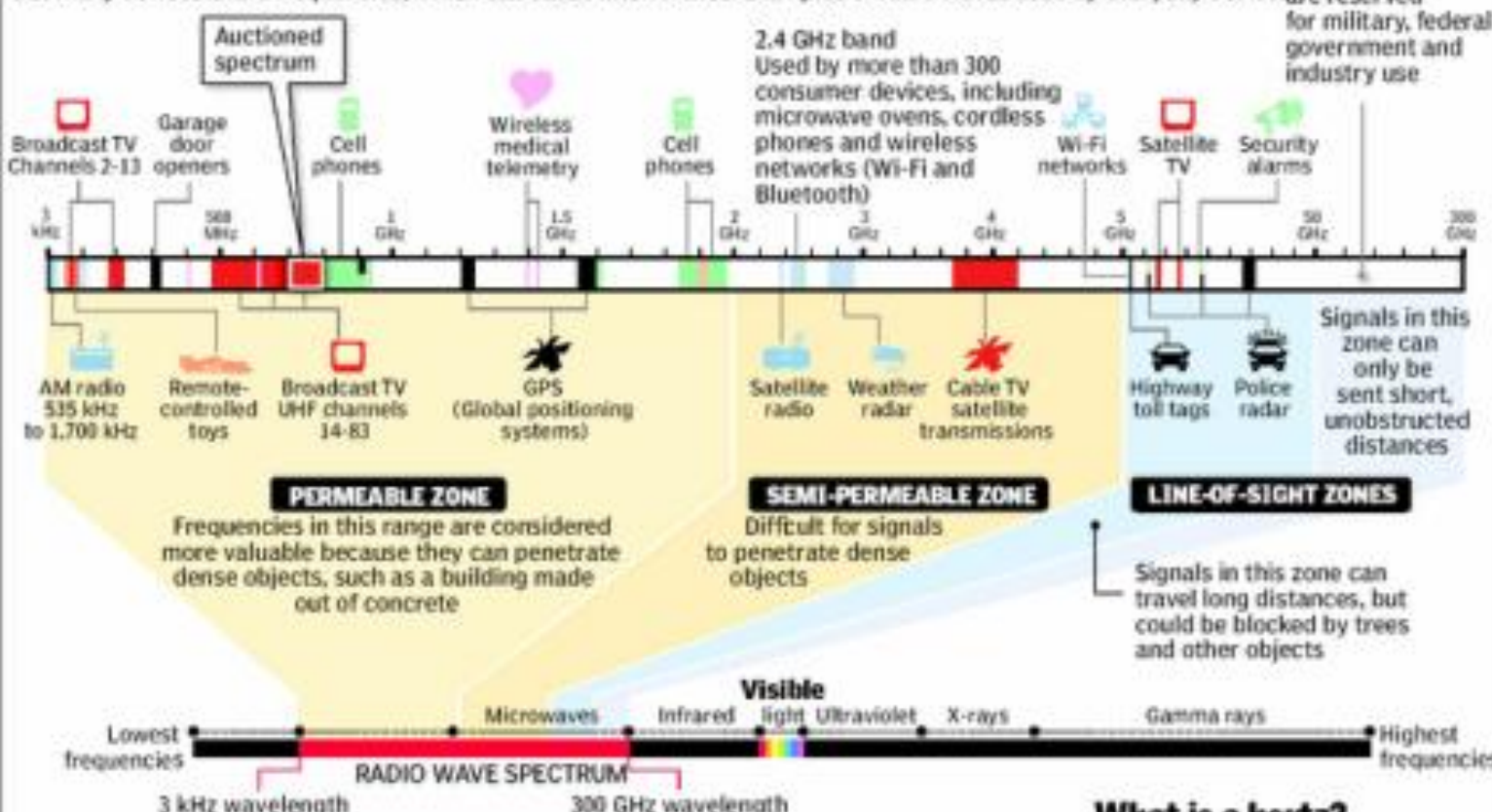
- ▶ Low-frequency signal: power/energy density spectrum concentrated around zero.
- ▶ High-frequency signal.
- ▶ Medium-frequency/ bandpass signal.

Bandwidth

- ▶ A quantitative measure that refers to the range of frequencies over which the power/energy density spectrum is concentrated.
- ▶ Narrowband, wideband, bandlimited.

Inside the radio wave spectrum

Almost every wireless technology – from cell phones to garage door openers – uses radio waves to communicate. Some services, such as TV and radio broadcasts, have exclusive use of their frequency within a geographic area. But many devices share frequencies, which can cause interference. Examples of radio waves used by everyday devices are reserved for military, federal government and industry use.



The electromagnetic spectrum

Radio waves occupy part of the electromagnetic spectrum, a range of electric and magnetic waves of different lengths that travel at the speed of light; other parts of the spectrum include visible light and x-rays; the shortest wavelengths have the highest frequency, measured in hertz.



What is a hertz?

One hertz is one cycle per second. For radio waves, a cycle is the distance from wave crest to crest.

1 kilohertz (kHz) = 1,000 hertz
1 megahertz (MHz) = 1 million hertz
1 gigahertz (GHz) = 1 billion hertz

Source: New America Foundation, MCT, Howstuffworks.com
Graphic: Nathaniel Levine, Sacramento Bee

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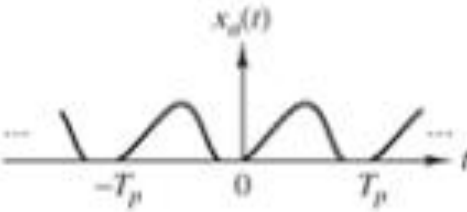
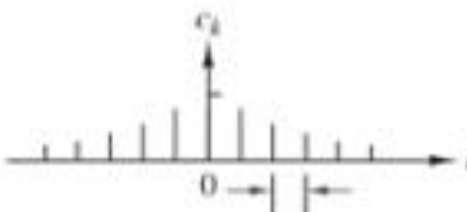
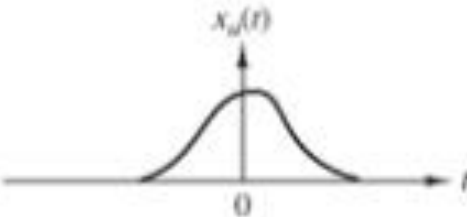
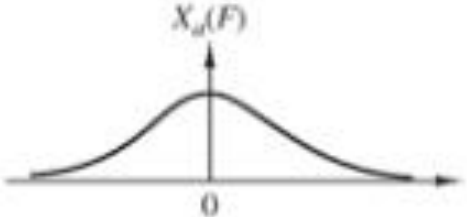
The frequency Ranges of Some Natural Signals

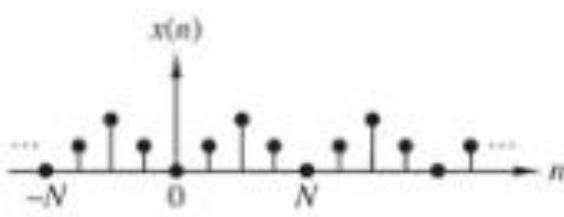
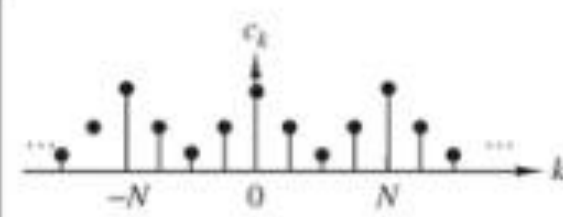
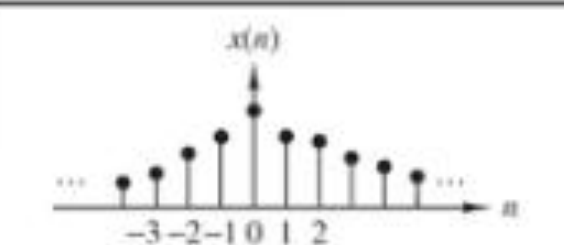
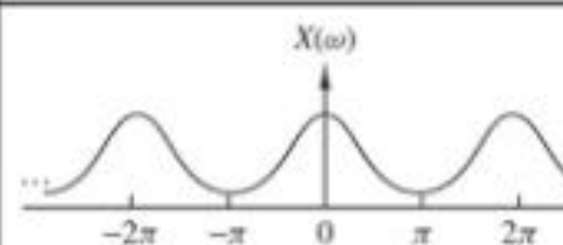
...

- ▶ Biological signals: Speech $100 - 4000\text{Hz}$, Sphygmomanogram $0 - 200\text{Hz}$
- ▶ Seismic Signals: Seismic exploration signals $10 - 100\text{Hz}$, Earthquakes $0.01 - 10\text{Hz}$
- ▶ Electromagnetic Signals: Infrared $3 \times 10^1 - 3 \times 10^{14}$, Bluetooth $2, 4002, 483.5\text{MHz}$

Frequency-Domain and Time-Domain Signal Properties

- ▶ Continuous-time $\Rightarrow$ aperiodic spectra
- ▶ Discrete-time $\Rightarrow$ periodic spectra
- ▶ Periodic signals $\Rightarrow$ discrete spectra
- ▶ Aperiodic (finite energy) $\Rightarrow$ continuous spectra

		Continuous-time signals	
		Time-domain	Frequency-domain
Periodic signals	Fourier series	 $c_k = \frac{1}{T_p} \int_{T_p} x_d(t) e^{-j2\pi k F_0 t} dt \Rightarrow$	 $F_0 = \frac{1}{T_p}$ $\Leftarrow x_d(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k F_0 t}$
		Continuous and periodic	Discrete and aperiodic
Aperiodic signals	Fourier transforms	 $X_d(F) = \int_{-\infty}^{\infty} x_d(t) e^{-j2\pi F t} dt \Rightarrow$	 $\Leftarrow x_d(t) = \int_{-\infty}^{\infty} X_d(F) e^{j2\pi F t} dF$
		Continuous and aperiodic	Continuous and aperiodic

		Discrete-time signals	
		Time-domain	Frequency-domain
Periodic signals	Fourier series	 $c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j(2\pi/N)kn}$	 $x(n) = \sum_{k=0}^{N-1} c_k e^{j(2\pi/N)kn}$
		Discrete and periodic	Discrete and periodic
Aperiodic signals	Fourier transforms	 $X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$	 $x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$
		Discrete and aperiodic	Continuous and periodic

Properties of the Fourier Transform for Discrete-Time Signals

Notation

Direct Transform (Analysis)

$$X(\omega) = F\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

Inverse Transform (Synthesis)

$$x(n) = F^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{2\pi} X(\omega)e^{j\omega n} d\omega$$

Fourier Transform pair

$$x(n) \xleftrightarrow{F} X(\omega)$$

Symmetry Properties of the Fourier Transform

Real Signals

- ▶ symmetry leads to simpler formulas for direct and inverse Fourier transform.
- ▶ If $x(n)$ is real, $x_I(n) = 0$ and using $e^{-j\omega} = \cos \omega - j \sin \omega$
- ▶ Spectrum of a Real signal has Hermitian symmetry:

$$X^*(\omega) = X(-\omega) \quad X_R(\omega) = \sum_{n=-\infty}^{\infty} x(n) \cos \omega n$$

$$X_I(\omega) = - \sum_{n=-\infty}^{\infty} x(n) \sin \omega n$$

- ▶ Magnitude has even symmetry $|X(\omega)| = |X(-\omega)|$
- ▶ Phase has odd symmetry $\angle X(-\omega) = -\angle X(\omega)$
- ▶ Table 4.4 for a summary of symmetry properties.

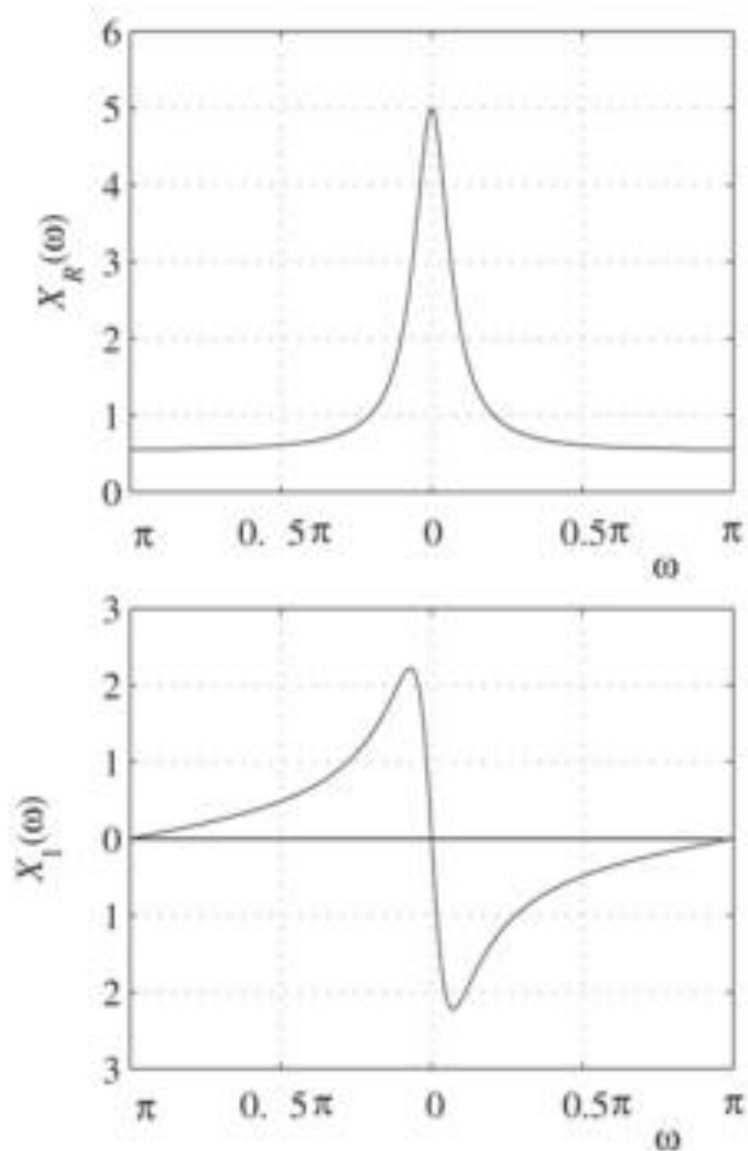


Figure 4.4.3 Graph of $X_R(\omega)$ and $X_I(\omega)$ for the transform in Example 4.4.1.

Proakis and Manolakis, *Digital Signal Processing*, Fourth Edition.
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Fourier Transform Theorems and Properties

Periodicity

- ▶ Discrete-time Fourier Transform is periodic with period 2π

$$X(\omega + 2\pi) = X(\omega)$$

Fourier Transform Theorems and Properties

Linearity

If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

Then

$$a_1 x_1(n) + a_2 x_2(n) \xleftrightarrow{F} a_1 X_1(\omega) + a_2 X_2(\omega)$$

Fourier Transform Theorems and Properties

Time Shifting

If

$$x(n) \xleftrightarrow{F} X(\omega)$$

Then

$$x(n - k) \xleftrightarrow{F} e^{-j\omega k} X(\omega)$$

$$F\{x(n - k)\} = |X(\omega)| e^{j[\angle X(\omega) - \omega k]}$$

- ▶ A shift in time domain by k samples, affects only the phase of the signal.

Fourier Transform Theorems and Properties

Frequency Shifting

If

$$x(n) \xleftrightarrow{F} X(\omega)$$

Then

$$e^{j\omega_0 n} x(n) \xleftrightarrow{F} X(\omega - \omega_0)$$

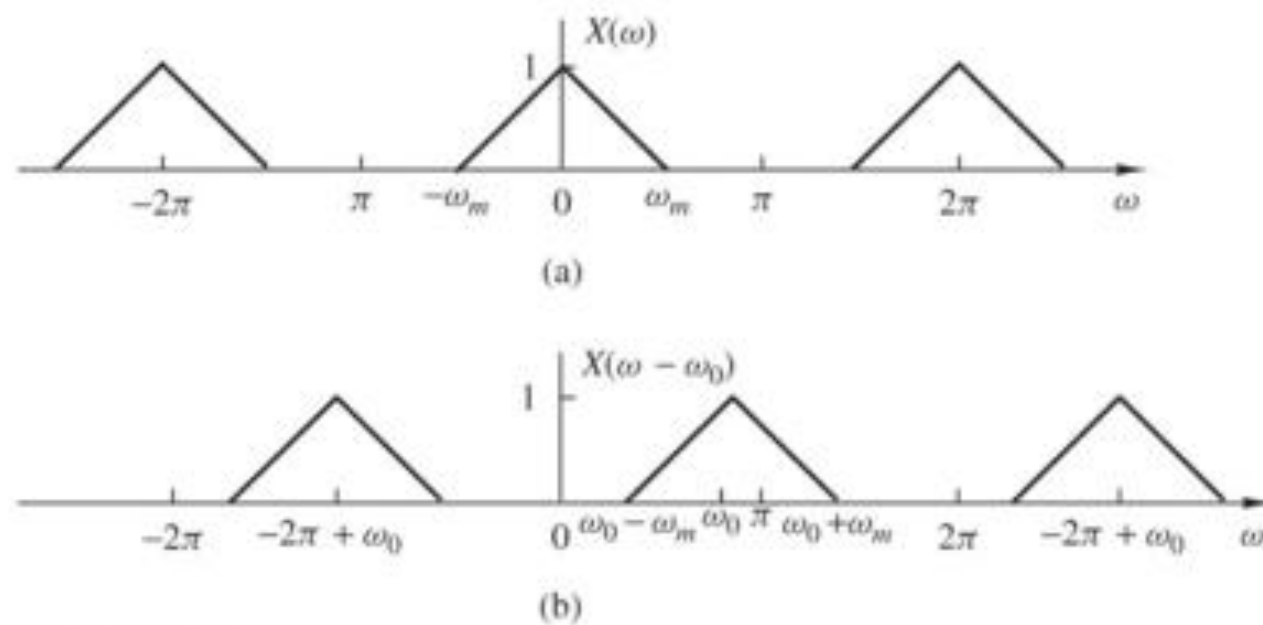


Figure 4.4.8 Illustration of the frequency-shifting property of the Fourier transform ($\omega_0 \leq 2\pi - \omega_m$).

Fourier Transform Theorems and Properties

Convolution

If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

Then

$$x(n) = x_1(n) * x_2(n) \xleftrightarrow{F} X(\omega) = X_1(\omega)X_2(\omega)$$

Fourier Transform Theorems and Properties

Modulation

If

$$x(n) \xleftrightarrow{F} X(\omega)$$

Then

$$x(n) \cos \omega_0 n \xleftrightarrow{F} \frac{1}{2} [X(\omega + \omega_0) + X(\omega - \omega_0)]$$

Proof uses $\cos \omega_0 n = \frac{1}{2} (e^{j\omega_0 n} + e^{-j\omega_0 n})$

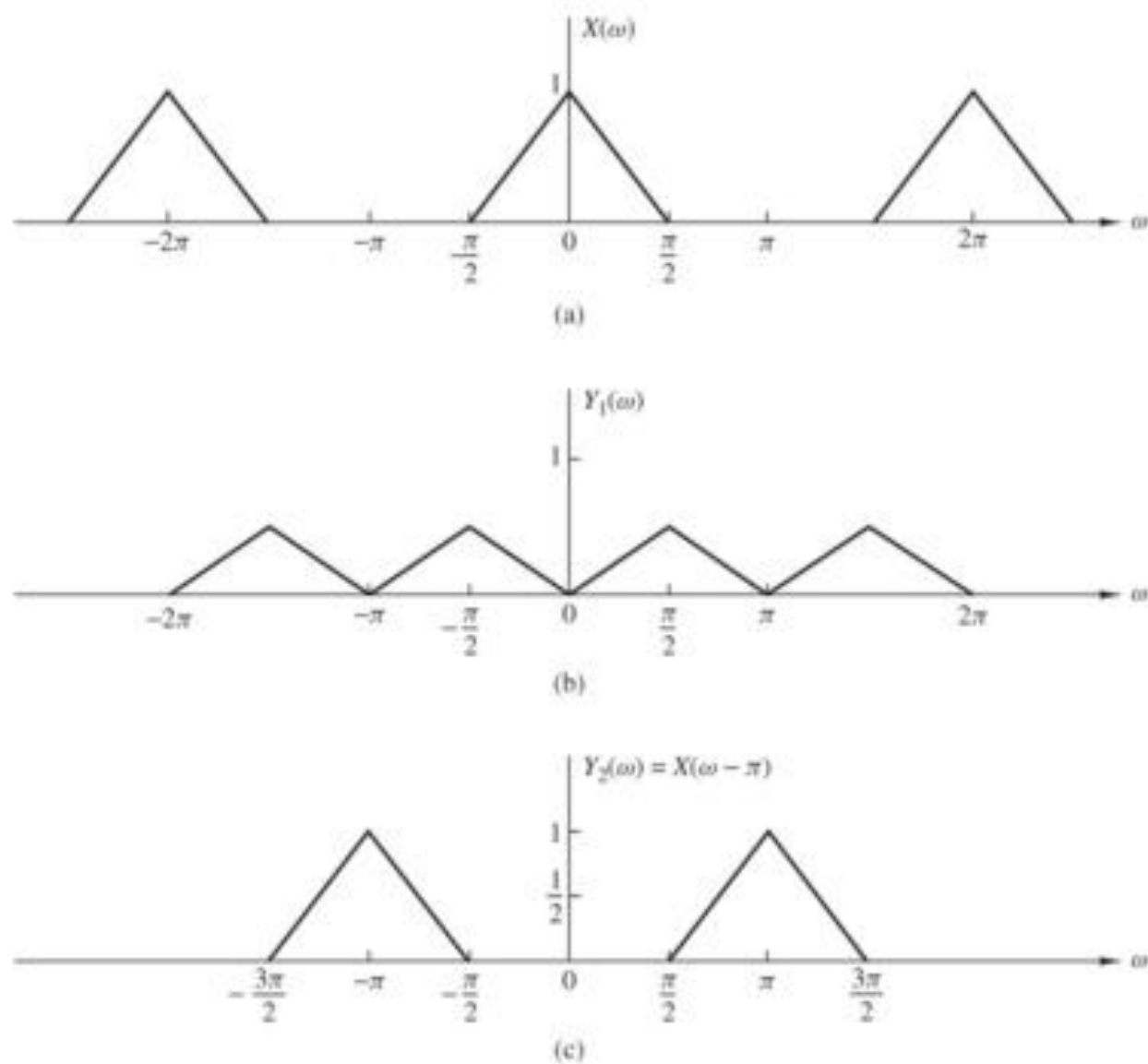


Figure 4.4.9 Graphical representation of the modulation theorem.

Fourier Transform Theorems and Properties

Correlation Theorem

If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

Then

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

Then

$$r_{x_1 x_2}(m) \xleftrightarrow{F} S_{x_1 x_2}(\omega) = X_1(\omega)X_2(-\omega)$$

Fourier Transform Theorems and Properties

Correlation Theorem: proof

cross correlation

$$r_{x_1 x_2}(n) = \sum_{-\infty}^{\infty} x_1(k) x_2(k - n)$$

multiply by $e^{-j\omega n}$ and then $\sum_{n=-\infty}^{\infty}$

$$\begin{aligned} S_{x_1 x_2}(\omega) &= \sum_{n=-\infty}^{\infty} r_{x_1 x_2}(n) e^{-j\omega n} = \\ &= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x_1(k) x_2(k - n) \right] e^{-j\omega n} \end{aligned}$$

Properties of the Fourier Transform for Discrete-Time Signals

Notation

- ▶ Direct Transform (Analysis)

$$X(\omega) = F\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \quad (1)$$

- ▶ Inverse Transform (Synthesis)

$$x(n) = F^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{2\pi} X(\omega)e^{j\omega n} d\omega \quad (2)$$

- ▶ Fourier Transform pair

$$x(n) \xleftrightarrow{F} X(\omega)$$

Symmetry Properties of the Fourier Transform

- Exploiting symmetry to arrive at simpler formulas for both the direct and inverse Fourier transform.

Suppose $x(n)$ and $X(\omega)$ are complex-valued functions.

$$\begin{aligned}x(n) &= x_R(n) + jx_I(n) \\X(\omega) &= X_R(\omega) + jX_I(\omega)\end{aligned}\tag{3}$$

using (remember $\cos -\omega = \cos \omega$ and $\sin -\omega = -\sin \omega$)

$$e^{-j\omega} = \cos \omega - j \sin \omega$$

Symmetry Properties of the Fourier Transform-cont

into equation 1

$$X_R(\omega) + jX_I(\omega) = \sum_{n=-\infty}^{\infty} [x_R(n) + jx_I(n)] [\cos \omega n - j \sin \omega n]$$

$$X_R(\omega) = \sum_{n=-\infty}^{\infty} [x_R(n) \cos \omega n + x_I(n) \sin \omega n]$$

$$X_I(\omega) = - \sum_{n=-\infty}^{\infty} [x_R(n) \sin \omega n - x_I(n) \cos \omega n]$$

Similarly with equation 1, we get

$$x_R(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [X_R(\omega) \cos \omega n - X_I(\omega) \sin \omega n] d\omega$$

$$x_I(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [X_R(\omega) \sin \omega n + X_I(\omega) \cos \omega n] d\omega$$

Special cases: Real signals

For a real $x(n) \Rightarrow x_I(n) = 0$

$$X_R(\omega) = \sum_{n=-\infty}^{\infty} x(n) \cos \omega n$$

$$X_I(\omega) = - \sum_{n=-\infty}^{\infty} x(n) \sin \omega n$$

Recall $\cos(-\omega n) = \cos \omega n$ and $\sin(-\omega n) = -\sin \omega n$ we get:

$$X_R(-\omega) = X_R(\omega), (\text{even})$$

$$X_I(-\omega) = -X_I(\omega), (\text{odd})$$

$$\Rightarrow X^*(\omega) = X(-\omega)$$

- The spectrum of a **real signal** has **Hermitian** symmetry.

Real Signals, cont.

- Magnitude and phase spectra for **real signals**

$$|X(\omega)| = \sqrt{X_R^2(\omega) + X_I^2(\omega)}, (\text{even})$$

$$\angle X(\omega) = \tan^{-1} \frac{X_I(\omega)}{X_R(\omega)}, (\text{odd})$$

- Inverse transform of a real signal, $x(n) = x_R(n)$ implies

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} [X_R(\omega) \cos \omega n - X_I(\omega) \sin \omega n] d\omega$$

$$x(n) = \frac{1}{\pi} \int_0^{\pi} [X_R(\omega) \cos \omega n - X_I(\omega) \sin \omega n] d\omega$$

Special cases: Real and even signals

- ▶ If $x(n)$ is real and even $x(-n) = x(n)$, then $x(n) \cos(\omega n)$ is **even** and $x(n) \sin(\omega n)$ is **odd**

$$X_R(\omega) = x(0) + 2 \sum_{n=1}^{\infty} x(n) \cos \omega n, \text{ (even)}$$

$$X_I(\omega) = 0$$

$$x(n) = \frac{1}{\pi} \int_0^{\pi} X_R(\omega) \cos \omega n d\omega$$

- ▶ real and even signals have a **real-valued spectra**, which is also even in ω .

Example 4.4.2

- Determine the Fourier transform of the signal

$$x(n) = \begin{cases} A, & -M \leq n \leq M \\ 0, & \text{elsewhere} \end{cases}$$

solution

Notice that $x(-n) = x(n) \Rightarrow x(n)$ real and even.

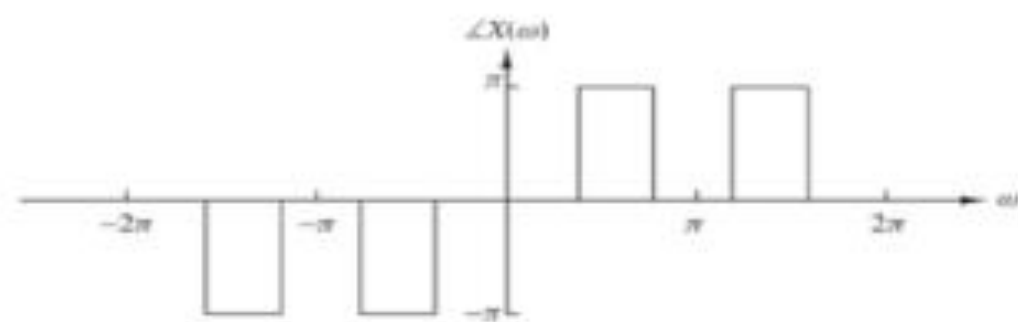
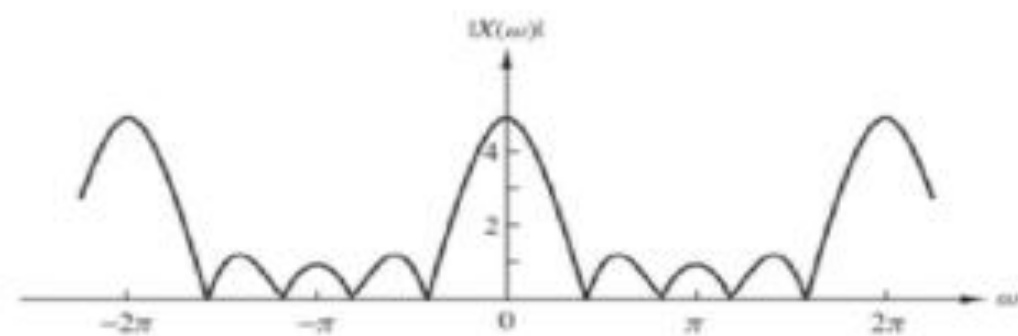
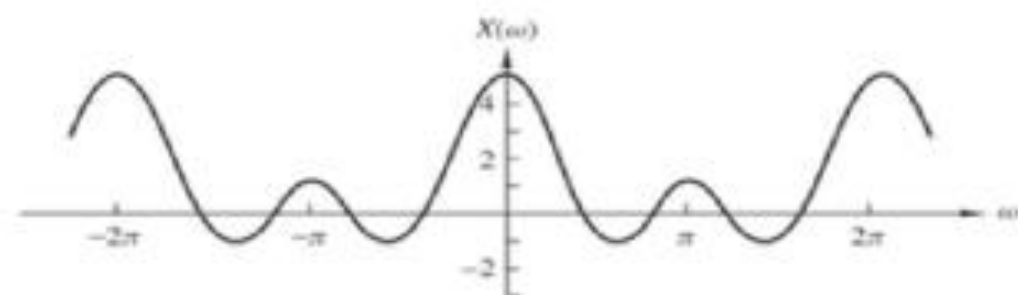
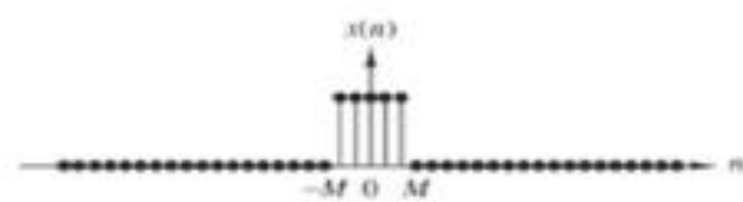
$$X(\omega) = X_R(\omega) = A \left(1 + 2 \sum_{n=1}^M \cos \omega n \right)$$

Since $X(\omega)$ is real,

$$|X(\omega)| = \left| A \left(1 + 2 \sum_{n=1}^M \cos \omega n \right) \right|$$

Example, cont.

$$\text{Phase}X(\omega) = \begin{cases} 0, & X(\omega) > 0 \\ \pi, & X(\omega) < 0 \end{cases}$$



Example: linearity

- Determine the Fourier transform of the signal

$$x(n) = a^{|n|}, -1 < a < 1$$

solution

Let $x(n) = x_1(n) + x_2(n)$ where

$$x_1(n) = \begin{cases} a^n, n \geq 0 \\ 0, n < 0 \end{cases}$$

and

$$x_2(n) = \begin{cases} a^{-n}, n < 0 \\ 0, n \geq 0 \end{cases}$$

example, cont.

we are going to use the linearity property $X(\omega) = X_1(\omega) + X_2(\omega)$ as follows:

$$X_1(\omega) = \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} = \sum_0^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Using geometric series rule for $|ae^{-j\omega n}| < 1$

$$X_1(\omega) = \frac{1}{1 - ae^{-j\omega}}$$

example, cont

Similarly, for $x_2(n)$

$$\begin{aligned}X_2(\omega) &= \sum_{n=-\infty}^{\infty} x_2(n) e^{-j\omega n} = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} \\&= \sum_{n=-\infty}^{-1} a e^{j\omega - n} = \sum_{k=1}^{\infty} (a e^{j\omega})^k \\&= \frac{a e^{j\omega}}{1 - a e^{j\omega}}\end{aligned}$$

Thus,

$$X(\omega) = X_1(\omega) + X_2(\omega)$$

example: convolution

- Determine the Fourier transform of the sequence

$$x_1(n) = x_2(n) = \{1, 1, 1\}$$

Solution

Since $x(n)$ is real and even,

$$X_1(\omega) = X_2(\omega) = 1 + 2 \cos \omega$$

Using the convolution property of the Fourier transform

$$\begin{aligned} X(\omega) &= X_1(\omega)X_2(\omega) = (1 + 2 \cos \omega)^2 \\ &= 3 + 4 \cos \omega + 2 \cos 2\omega \\ &= 3 + 2(e^{j\omega} + e^{-j\omega}) + (e^{j2\omega} + e^{-j2\omega}) \end{aligned}$$

$$x(n) = \{1, 2, 3, 2, 1\}$$

Properties of the Fourier Transform for Discrete-time signals

Wiener-Khintchine theorem

For $x(n)$, a real signal

$$r_{xx}(l) \xleftrightarrow{F} S_{xx}(\omega)$$

- ▶ Energy spectral density of an energy signal is the Fourier transform of the signal autocorrelation sequence.
- ▶ Recall, $r_{x_1 x_2}(m) \xleftrightarrow{F} S_{x_1 x_2}(\omega) = X_1(\omega)X_2(-\omega)$

Properties of the Fourier Transform for Discrete-time signals

Parseval's theorem

If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

And

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

Then

$$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega)d\omega$$

Properties of the Fourier Transform, Parseval's theorem

- Proof: starting with the right side

$$\begin{aligned} & \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} \right] X_2^*(\omega) d\omega \\ &= \sum_{n=-\infty}^{\infty} x_1(n) \frac{1}{2\pi} \int_{-\infty}^{\infty} X_2^*(\omega) e^{-j\omega n} d\omega = \sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) \end{aligned}$$

- For $x(n) = x_1(n) = x_2(n)$, Parseval's reduces to:

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Properties of the Fourier Transform, Parseval's theorem

- ▶ Can use Parseval's theorem to find:

$$E_x = r_{xx}(0) = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{xx}(\omega) d\omega$$

- ▶ Example: refer back to 4.4.4 $E_x = 3$, and $r_{xx}(0) = 3$

Differentiation in the frequency domain

$$\begin{aligned}x(n) &\xleftrightarrow{\text{F}} X(\omega) \\ nx(n) &\xleftrightarrow{\text{F}} j \frac{dX(\omega)}{d\omega}\end{aligned}$$

► Proof:

$$\begin{aligned}\frac{dX(\omega)}{d\omega} &= \frac{d}{d\omega} \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right] \\ &= \sum_{n=-\infty}^{\infty} x(n) \frac{d}{d\omega} e^{-j\omega n} \\ &= -j \sum_{n=-\infty}^{\infty} nx(n) e^{-j\omega n}\end{aligned}$$